

DECODELABS INTERNSHIP

Introduction to IoT (Internet of Things)

The **Internet of Things (IoT)** is a network of physical devices connected to the internet that can collect, share, and exchange data automatically. These devices use sensors, software, and communication technologies to interact with other devices and systems.

Examples of IoT Devices

- Smart home lights
- Smart watches
- Smart thermostats
- Connected vehicles
- Smart agriculture systems

Basic IoT Architecture

IoT architecture is generally divided into three main layers:

1. **Devices Layer (Sensors & Actuators)**
2. **Network Layer (Wi-Fi, Bluetooth, Cellular, etc.)**
3. **Cloud Layer (Data Storage & Processing)**

1. Devices Layer

This layer contains the physical devices.

Components:

- **Sensors:** Collect data from the environment
 - Temperature sensor
 - Humidity sensor
 - Motion sensor
- **Actuators:** Perform actions based on commands
 - Motor
 - Relay
 - LED

Example: A temperature sensor measures room temperature and sends the reading.

2. Network Layer

This layer transfers data between devices and the cloud.

Communication Technologies:

- Wi-Fi
- Bluetooth
- Zigbee
- LoRaWAN
- Cellular (4G/5G)
- Ethernet

Functions:

- Data transmission
- Device connectivity
- Security and routing

3. Cloud Layer

Data Flow in an IoT System

Example: Smart Temperature Monitoring System

1. Sensor measures temperature
2. Microcontroller (ESP32) reads the sensor data
3. Data is sent through Wi-Fi
4. Data reaches the cloud server
5. Cloud stores and analyses the data
6. User views data on a mobile app
7. If temperature exceeds a limit, the cloud sends a command
8. An actuator (fan) turns ON automatically

Flow Diagram: Temperature Sensor → ESP32 → Wi-Fi → Cloud Server → Mobile Application → Fan ON

Real-World IoT Use Cases

1. Smart Home

- Smart lights
- Smart locks
- Smart thermostats
- Voice assistants
- **Benefits:** Convenience, energy saving, remote control

2. Healthcare

- Fitness trackers
- Heart-rate monitoring devices
- Remote patient monitoring
- **Benefits:** Continuous health tracking, early detection of problems

3. Smart Agriculture

- Soil moisture sensors
- Automated irrigation systems
- Weather monitoring
- **Benefits:** Water conservation, increased crop yield

4. Smart Cities

- Smart traffic management
- Smart street lighting
- Waste management systems
- **Benefits:** Reduced congestion, better resource utilization

5. Industrial IoT (IIoT)

- Machine monitoring
- Predictive maintenance
- Asset tracking
- **Benefits:** Reduced downtime, increased productivity

Key Skills and Concepts

IoT Fundamentals

- Sensors and actuators
- Microcontrollers (ESP32, Arduino, Raspberry Pi)
- Communication protocols
- Cloud computing

System Understanding

- Device-to-device communication
- Device-to-cloud communication
- Data collection and processing
- Security and privacy

Conclusion

IoT connects physical devices to the internet so they can collect and exchange data. An IoT system consists of:

1. Devices (sensors and actuators)
2. Network (communication technologies)
3. Cloud (storage and processing)

Data flows from **sensors** → **network** → **cloud** → **user/application**, enabling automation and smart decision-making in areas such as homes, healthcare, agriculture, cities, and industries.